THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** metre	standard unit
2	**(d)** a ruler (scale)	measuring tool
3	**(d)** the zero mark (or a clear mark) of the scale	correct measuring
4	**(c)** handspans, foot-lengths and cubits	non-standard units
5	**(d)** linear motion	straight line
6	**(c)** random	unpredictable
7	**(a)** linear	straight line
8	**(a)** at rest with respect to the bus stop	relative to reference
9	**(b)** 2500 m	×1000
10	**(c)** a thread and scale	curved line

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	So that measurements are the same everywhere and can be compared.	1+1
12	Metre; kilometre.	1+1
13	350 cm.	1+1
14	4.5 km.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place the zero mark at one edge (1); keep the eye directly above the other edge (1); read the mark (1).	1+1+1
16	Place a thread along the curve (1); mark the end points (1); stretch and measure on a scale (1).	1+1+1
17	Rotation: turning about an axis, e.g. fan (1.5); oscillation: to-and-fro, e.g. swing (1.5).	1½+1½
	OR: Linear: car on a straight road (1); circular: stone tied to a string (1); periodic: pendulum (1).	1+1+1
18	2,350 m (1.5); 2,35,000 cm (1.5).	1½+1½
	OR: Zero of scale at start (1); eye straight above (1); avoid broken scales (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Reason (1). (ii) mm, cm, m, km relations (2). (iii) 5,600 m; 5,60,000 cm (2).	1+2+2
	OR: (i) Steps (2). (ii) Parallax, wrong zero, broken scale (2). (iii) Remedies (1).	2+2+1
20	(i) Different hand sizes (1). (ii) Handspan (1). (iii) Use a ruler or metre scale (1). (iv) Standard units (2).	1+1+1+2
	OR: (i) Procedure (2). (ii) 37 km (2). (iii) Coastline length (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 5: **Measurement of Length and Motion** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 100 centimetres	conversion
2	**(c)** thread and a ruler	method
3	**(c)** kilometre	large distances
4	**(b)** handspans are not standard	non-standard
5	**(b)** rotational (circular) motion	turns about a fixed axis
6	**(b)** periodic (circular/orbital)	repeats yearly
7	**(c)** rotation	about an axis
8	**(c)** the train	relative rest
9	**(b)** distance moved in unit time	definition
10	**(b)** 1 millimetre	least count

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Error due to viewing the scale at an angle; look straight above the mark.	1+1
12	Lay a thread along the curve, then straighten it and measure with a scale.	1+1
13	A fixed point with respect to which position is described.	1+1
14	Change in position of an object with time.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	They differ from person to person (1.5); results cannot be compared (1.5).	1½+1½
16	Choose a reference point and give distance and direction (3).	1+2
17	At rest relative to the bus; in motion relative to the road (3).	1+2
	OR: Periodic: clock hands, pendulum, Earth (1.5); random: bird flying, leaf falling, children playing (1.5).	1½+1½
18	0.36 m (3).	1+2
	OR: Standard and accurate (3).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Linear, circular, rotational, oscillatory, random (3). (ii) Periodic (1). (iii) Random (1).	3+1+1
	OR: (i) Oscillatory (1). (ii) Yes (1). (iii) 2 s (1). (iv) Use a stopwatch for many swings and divide (2).	1+1+1+2
20	(i) Fixed place (1). (ii) Depends on reference (2). (iii) A passenger moves relative to platform but is at rest relative to the train (2).	1+2+2
	OR: (i) 275 cm; 2,750 mm (2). (ii) 46 m; 0.046 km (2). (iii) 4 m 25 cm (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.